LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR SOFT, PROCESSABLE, AND LOW-VISCOSITY NITRILE RUBBER SYSTEMS

1. ABSTRACT

The supplied patent set addresses two closely related but distinct low-acrylonitrile nitrile rubber directions: (1) soft, carboxylated nitrile-butadiene latex formulations for dipped articles with natural-rubber-like feel, and (2) low molecular weight hydrogenated nitrile rubber (HNBR) made by post-polymerization chain scission via metathesis to improve processability. Across the glove-focused patents, the target material is a carboxylated NBR latex blend containing two acrylonitrile terpolymers, with total acrylonitrile content between 17 and 45 wt%, methacrylic acid below 15 wt%, and butadiene as the balance. The blend ratio of the two terpolymers is 60:40 to 40:60, with the first terpolymer having lower modulus than the second. The formulation is tuned with ionic crosslinking components (alkali agent plus metallic oxide) and optional sulfur/accelerator packages, and the latex pH is maintained at 9 to 12.5. The process emphasis is on dipped elastomeric articles, especially gloves, where lower modulus and reduced thickness are used to mimic natural rubber force-strain response while retaining nitrile's chemical resistance.

The HNBR patents focus on a different route to low-viscosity nitrile-based elastomers: a nitrile rubber precursor containing at least one conjugated diene and at least one α,β-unsaturated nitrile, optionally with unsaturated carboxylic comonomers, is subjected to metathesis in the presence of Ru- or Os-based catalysts, optionally with a co-olefin and solvent. The claimed product window is MW 30,000 to 250,000, Mooney viscosity 3 to 50, and MWD below 2.5. The synthesis narrative highlights controlled molecular-weight reduction rather than microstructure alteration, contrasting with older mastication or acid degradation methods. The extracted examples indicate solution processing in methylcyclobenzene at 6–15 wt% solids, use of 1-hexene as co-olefin, and reactor conditions around 80 °C, with catalyst loadings down to 0.05 phr. Overall, the landscape shows two major strategies for low-acrylonitrile NBR performance: formulation-level softening for latex articles and catalyst-driven molecular-weight reduction for HNBR feedstocks and shaped articles.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8470422B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013-06-25
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example EXAMPLE 12:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011-04-05
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Concentration: 6 % (6 % of Concentration) - Concentration: 150 g (150 g of Concentration) - Temperature: 0.05 phr (temperature ... 0.05 phr) - Rubber: 200 g (200 g of Rubber) - Mcb: 1133 g (1133 g of Mcb) - Rubber: 200 g (200 g of Rubber) - Mcb: 1133 g (1133 g of Mcb) - Rubber: 75 g (75 g of Rubber) - Mcb: 1175 g (1175 g of Mcb) - Rubber: 75 g (75 g of Rubber) - Mcb: 1175 g (1175 g of Mcb) - 1-hexene: 150 g (150 g of 1-hexene) - 15: 15 % (15 % of 15)

Relevant Experimental Evidence - Example EXAMPLE 3: 15: 15 %, Concentration: 6 %, Concentration: 150 g, Temperature: 0.05 phr - Example EXAMPLE 4: Rubber: 75 g, Mcb: 1175 g, 1-hexene: 150 g - Example EXAMPLE 1: Rubber: 200 g, Mcb: 1133 g - Example EXAMPLE 2: Rubber: 200 g, Mcb: 1133 g - Example EXAMPLE 3: Rubber: 75 g, Mcb: 1175 g

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011-05-31
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Concentration: 6 % (6 % of Concentration) - Concentration: 150 g (150 g of Concentration) - Temperature: 0.05 phr (temperature ... 0.05 phr) - Rubber: 200 g (200 g of Rubber) - Mcb: 1133 g (1133 g of Mcb) - Rubber: 200 g (200 g of Rubber) - Mcb: 1133 g (1133 g of Mcb) - Rubber: 75 g (75 g of Rubber) - Mcb: 1175 g (1175 g of Mcb) - Rubber: 75 g (75 g of Rubber) - Mcb: 1175 g (1175 g of Mcb) - 1-hexene: 150 g (150 g of 1-hexene) - 15: 15 % (15 % of 15)

Relevant Experimental Evidence - Example EXAMPLE 3: 15: 15 %, Concentration: 6 %, Concentration: 150 g, Temperature: 0.05 phr - Example EXAMPLE 4: Rubber: 75 g, Mcb: 1175 g, 1-hexene: 150 g - Example EXAMPLE 1: Rubber: 200 g, Mcb: 1133 g - Example EXAMPLE 2: Rubber: 200 g, Mcb: 1133 g - Example EXAMPLE 3: Rubber: 75 g, Mcb: 1175 g

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US9926439B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018-03-27
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example EXAMPLE 12:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US8470422B2 and US9926439B2 are closely aligned: both claim carboxylated nitrile-butadiene latex formulations with acrylonitrile content of 17–45 wt%, methacrylic acid below 15 wt%, and butadiene as the balance; both use a 60:40 to 40:60 blend of two acrylonitrile terpolymers, with the first polymer lower in modulus than the second.
- Both glove patents emphasize ionic/covalent crosslink balance rather than polymer synthesis changes: 0–1.5 parts alkali agent plus 0.5–1.5 parts metallic oxide, with total >1.0 part per 100 parts rubber, and optional 0.5–1.5 parts sulfur plus 0.5–1.5 parts accelerator; pH is maintained at 9 to 12.5.
- The glove patents are process/formulation patents for dipped articles, while US7919563B2 and US7951875B2 are post-polymerization modification patents for HNBR; the latter target lower MW, lower Mooney viscosity, and narrower MWD rather than glove feel.
- US7919563B2 and US7951875B2 share the same core HNBR metathesis concept and product window: MW 30,000–250,000, Mooney 3–50, and MWD <2.5, using Ru/Os metathesis catalysts with optional co-olefin and solvent.
- The HNBR patents differ mainly in claim scope: US7919563B2 claims hydrogenated nitrile rubber broadly, while US7951875B2 specifically claims hydrogenated carboxylated nitrile rubber and explicitly recites unsaturated mono- or dicarboxylic acids, esters, or amides thereof.
- The extracted examples for the HNBR patents indicate solution concentrations of 6 wt% and 15 wt%, 1-hexene as co-olefin, and reactor temperature around 80 °C; catalyst loadings in the extracted data include 0.05 phr.
- Historically, the HNBR patents represent a controlled depolymerization/metathesis approach to viscosity reduction, whereas the glove patents represent a newer formulation strategy to tune softness without abandoning conventional nitrile crosslink chemistry.
- Across the set, recurring themes are preservation of nitrile chemical resistance, reduction of stiffness or viscosity, and reliance on controlled composition/process windows rather than extreme additive levels or uncontrolled degradation.

4. CONCLUSION

The evidence supports two practical synthesis routes for low-acrylonitrile NBR-derived materials. For soft latex articles, the best-supported parameters are a carboxylated NBR blend containing 17–45 wt% acrylonitrile, less than 15 wt% methacrylic acid, and a 60:40 to 40:60 blend of two terpolymers with different modulus, combined with modest ionic crosslinking (alkali agent plus metallic oxide) and optional sulfur/accelerator cure at pH 9–12.5. For low-viscosity HNBR feedstocks, the patents support controlled metathesis of NBR using Ru- or Os-based catalysts, optionally with a co-olefin such as 1-hexene and solvent, to reach MW 30,000–250,000, Mooney 3–50, and MWD below 2.5 without changing microstructure. Overall, the strongest recommendation from the patent set is to use composition blending and balanced crosslinking for glove softness, and catalyst-mediated molecular-weight reduction for processable HNBR, rather than relying on harsh mastication or acid degradation.

5. REFERENCES

- US8470422B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2013 | https://patents.google.com/patent/US8470422B2/en
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | https://patents.google.com/patent/US7919563B2/en
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | https://patents.google.com/patent/US7951875B2/en
- US9926439B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2018 | https://patents.google.com/patent/US9926439B2/en